Track of a Line Segment on a Rotating Rectangle

Kinematic Analysis & Coordinate Transformation

Classification: Internal

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Objective: Derive the parametric trajectory of a line segment mounted on a rotating rectangle, subject to a secondary pivot rotation about its top endpoint. Compute the homogeneous transformation matrix from the world frame to a body-fixed target frame attached to the segment center.

Contents

1 Problem Setup

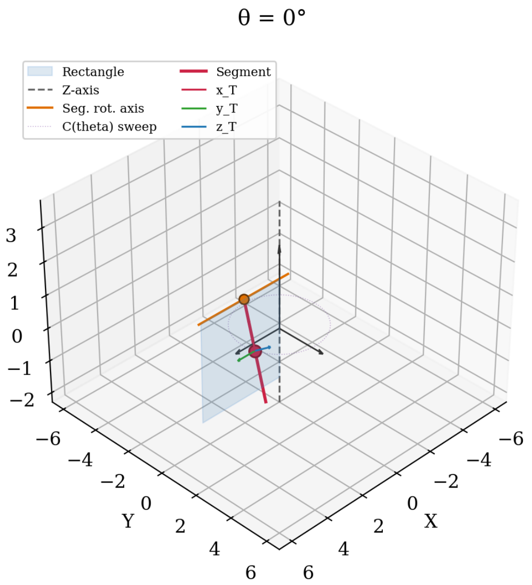
A rectangle rotates about its left vertical edge, which coincides with the z-axis. A line segment (perpendicular to the rectangle plane, i.e. parallel to the z-axis) is attached to the rectangle. The segment then undergoes a second rotation around an axis passing through its top endpoint.

- The rectangle's rotation edge is the z-axis $(0, 0, z)$.
- At $\theta = 0$ the rectangle lies in the xz-plane; the segment center is at $(r,\ 0,\ h)$, at distance r from the z-axis.
- The rectangle rotates by an angle θ about the z-axis.
- h is the height of the center point of the segment.
- Let the segment have half-length L (so its full length is $2L$).
- The segment rotates by an angle ϕ about an axis passing through its *top* endpoint, with unit direction vector $\mathbf{u} = (\cos\theta,\ \sin\theta,\ 0)$ (the direction of the rectangle plane at angle θ).

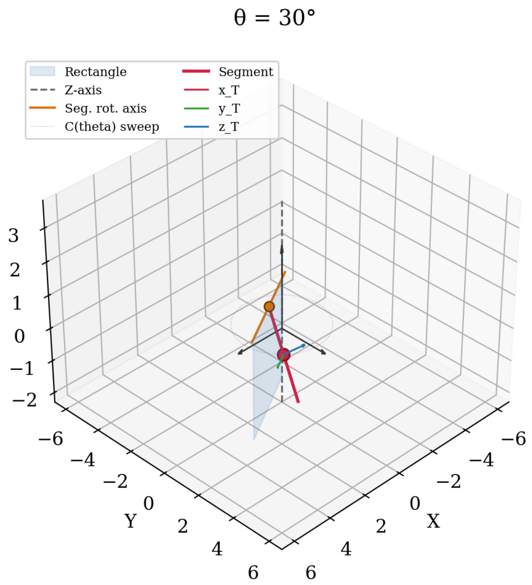
We seek the parametric equation of the *center point* of the segment after both rotations, as a function of θ and ϕ.

2 Visual Overview

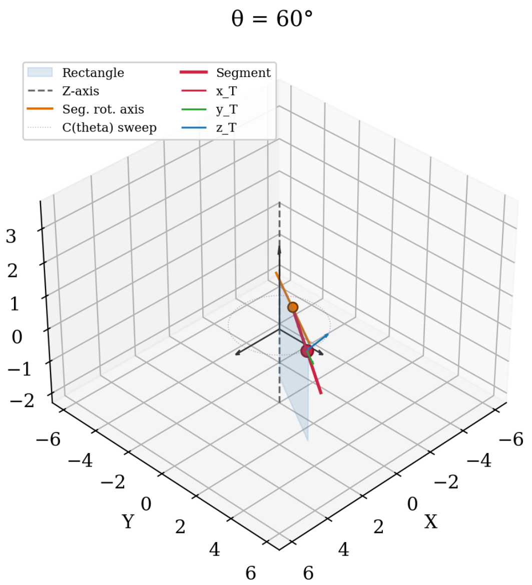
Figure 1 shows the configuration at four values of θ with $\phi = 24°$ fixed ($r = 2$, $L = 1.5$, $h = 0$).



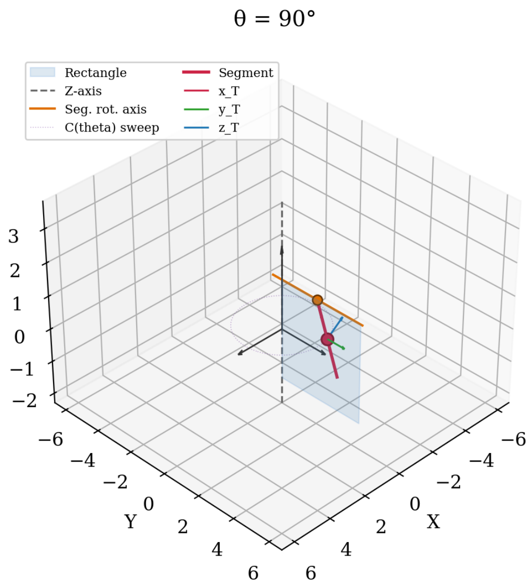
(a) $\theta = 0°$



(b) $\theta = 30°$



(c) $\theta = 60°$



(d) $\theta = 90°$

Figure 1: Configuration at $\phi = 24°$ for four values of θ. The blue dotted curve is the θ-sweep of $C(\theta, \phi)$; the target frame $\{\hat{\mathbf{x}}_T, \hat{\mathbf{y}}_T, \hat{\mathbf{z}}_T\}$ is shown at the segment center.

3 Numerical Verification

Table 1 lists $C(\theta, \phi)$ at 16 (θ, ϕ) combinations, computed by the script with $r = 2$, $L = 1.5$, $h = 0$. These values satisfy all five verification cases from Section 8.

Table 1: Center point $C(\theta, \phi)$ at selected angles.

θ	ϕ	C_x	C_y	C_z	Checks
0°	0°	2.0000	0.0000	0.0000	$\phi = 0$: $C_z = h$
0°	45°	2.0000	1.0607	0.4393	$\theta = 0$: $C_x = r$
0°	90°	2.0000	1.5000	1.5000	$\phi = 90°$: $C_z = h + L$
0°	180°	2.0000	0.0000	3.0000	$\phi = 180°$: $C_z = h + 2L$
30°	0°	1.7321	1.0000	0.0000	$\lvert(C_x, C_y)\rvert = r = 2$
30°	45°	1.2017	1.9186	0.4393	
30°	90°	0.9821	2.2990	1.5000	
30°	180°	1.7321	1.0000	3.0000	$C(\theta, \pi) = P(\theta) + (0, 0, 2L)$
60°	0°	1.0000	1.7321	0.0000	$\lvert(C_x, C_y)\rvert = r = 2$
60°	45°	0.0814	2.2624	0.4393	
60°	90°	−0.2990	2.4821	1.5000	
60°	180°	1.0000	1.7321	3.0000	
90°	0°	0.0000	2.0000	0.0000	$\theta = 90°$: $C_x = 0$
90°	45°	−1.0607	2.0000	0.4393	
90°	90°	−1.5000	2.0000	1.5000	
90°	180°	−0.0000	2.0000	3.0000	

4 Function Surface

Figure 2 shows the 2-parameter surface $C(\theta, \phi)$ as both θ and ϕ vary over $[0, 2\pi]$, with $r = 2$, $L = 1.5$, $h = 0$. The surface is a torus-like shape: θ sweeps a circle of radius r around the z-axis, modulated by a sinusoidal ripple of amplitude $L \sin \phi$ from the segment tilt. The right panel shows the XY projection at fixed ϕ values: at $\phi = 0$ the curve is a circle of radius r; as ϕ increases the curve flattens into a nephroid-like shape with maximum radial excursion $r + L$.

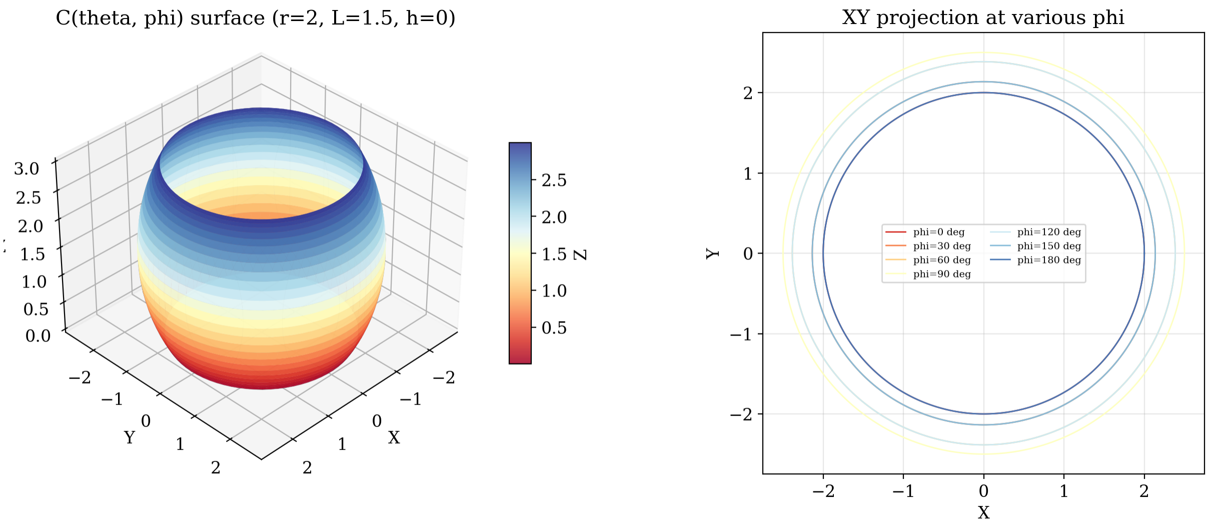


Figure 2: $C(\theta, \phi)$ parametric surface (left) and XY projection at selected ϕ values (right).

5 Reference Positions

5.1 Center point after rectangle rotation only

At $\theta = 0$ the segment center is at $(r, 0, h)$. Rotating counterclockwise by θ about the z-axis:

$$P(\theta) = \big(r\cos\theta,\ r\sin\theta,\ h\big). \tag{1}$$

The distance from this point to the z-axis is $\sqrt{(r\cos\theta)^2 + (r\sin\theta)^2} = r$, consistent with the definition of r.

5.2 Top endpoint

Since the segment is vertical (parallel to the z-axis), the top endpoint is displaced by $+L$ from the center:

$$T(\theta) = P(\theta) + (0,\ 0,\ L) = \big(r\cos\theta,\ r\sin\theta,\ h + L\big). \tag{2}$$

5.3 Vector from top to center

$$\mathbf{d} = P - T = (0,\ 0,\ -L). \tag{3}$$

6 Derivation via Rodrigues' Rotation Formula

The segment rotates by angle ϕ around the axis through T with unit direction $\mathbf{u} = (\cos\theta,\ \sin\theta,\ 0)$. Let P' be the new position of the center point after this rotation.

Rodrigues' formula for rotating a vector $\mathbf{d}$ by angle ϕ about a unit axis $\mathbf{u}$ is

$$\mathbf{R}_{\mathbf{u}}(\phi)\,\mathbf{d} = \mathbf{d}\cos\phi + (\mathbf{u}\times\mathbf{d})\sin\phi + \mathbf{u}\,(\mathbf{u}\cdot\mathbf{d})\,(1-\cos\phi). \tag{4}$$

Since $P' = T + \mathbf{R}_{\mathbf{u}}(\phi)\,\mathbf{d}$, we compute each term.

6.1 Dot product term

$$\mathbf{u}\cdot\mathbf{d} = (\cos\theta)(0) + (\sin\theta)(0) + (0)(-L) = 0.$$

The projection term vanishes.

6.2 Cross product term

$$\mathbf{u}\times\mathbf{d} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \cos\theta & \sin\theta & 0 \\ 0 & 0 & -L \end{vmatrix} = \big(-L\sin\theta,\ L\cos\theta,\ 0\big).$$

6.3 Assembling the result

$$
\begin{aligned}
P' &= T + \mathbf{d}\cos\phi + (\mathbf{u}\times\mathbf{d})\sin\phi + \mathbf{u}\,(\mathbf{u}\cdot\mathbf{d})\,(1-\cos\phi) \\
&= T + (0,0,-L)\cos\phi + (-L\sin\theta,\ L\cos\theta,\ 0)\sin\phi + \mathbf{0} \\
&= \begin{pmatrix} r\cos\theta \\ r\sin\theta \\ h+L \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ -L\cos\phi \end{pmatrix} + \begin{pmatrix} -L\sin\theta\sin\phi \\ L\cos\theta\sin\phi \\ 0 \end{pmatrix}.
\end{aligned}
\tag{5}
$$

7 Result

The track of the center point $C(\theta, \phi)$ is the parametric surface:

$$
\boxed{C(\theta,\phi) = \begin{pmatrix} r\cos\theta - L\sin\theta\,\sin\phi \\ r\sin\theta + L\cos\theta\,\sin\phi \\ h + L\,(1-\cos\phi) \end{pmatrix}}.
\tag{6}
$$

Table 2: Parameter definitions

Symbol	Meaning
θ	Rotation angle of the rectangle about the z-axis
ϕ	Rotation angle of the segment about its top-end axis
r	Distance from segment to the z-axis
h	Height of the segment center point
L	Half-length of the segment

8 Verification

Case 1: $\phi = 0$ (no segment rotation)

$$C(\theta, 0) = \big(r\cos\theta,\ r\sin\theta,\ h\big) = P(\theta).$$

The center returns to its position on the rotating rectangle.

Case 2: $\phi = \pi$ (segment flipped 180°)

$$C(\theta, \pi) = \big(r\cos\theta,\ r\sin\theta,\ h + 2L\big).$$

The center moves to a point L *above* the original top endpoint, consistent with a half-turn about the top.

Case 3: $\phi = \pi/2$ (segment horizontal)

The z-coordinate becomes $h + L$, and the horizontal displacement has magnitude L in the direction perpendicular to $\mathbf{u}$. This corresponds to the segment lying flat, perpendicular to the plane containing $\mathbf{u}$ and the z-axis.

Case 4: $\theta = 0$ (rectangle in xz-plane)

$$C(0, \phi) = \big(r,\ L\sin\phi,\ h + L(1 - \cos\phi)\big).$$

The center traces a circle of radius L in the yz-plane, centered at $(r,\,0,\,h+L)$, which is exactly the top endpoint of the segment at $\theta = 0$.

Case 5: distance preservation

For any θ, the distance from $C(\theta,\phi)$ to the z-axis varies between $|r - L\sin\phi|$ and $r + L\sin\phi$, never exceeding $r + L$. At $\phi = \pi/2$, the segment lies flat and the center is at its farthest horizontal reach.

9 Interpretation

For a fixed θ, the curve traced by varying ϕ is a circle of radius L in the plane perpendicular to the axis $\mathbf{u} = (\cos\theta, \sin\theta, 0)$. The center of this circle is at $T(\theta)$ — the top endpoint of the segment.

For a fixed ϕ, varying θ traces a curve on a cylindrical surface of radius $\approx r$ around the z-axis, with a sinusoidal ripple of amplitude $L\sin\phi$ from the segment tilt. The z-coordinate depends only on ϕ, so horizontal slices at constant ϕ all lie in the same z-plane.

10 Transformation from World to Target Frame

10.1 Target frame definition

Attach a right-handed coordinate frame $\{T\}$ to the segment:

- **Origin**: the segment center point $C(\theta,\phi)$.
- $\hat{\mathbf{x}}_T$: along the segment, pointing from bottom to top. Obtained by rotating the unit vertical $(0,0,1)$ by ϕ about $\mathbf{u}$.
- $\hat{\mathbf{y}}_T$: parallel to the top-end rotation axis $\mathbf{u} = (\cos\theta,\ \sin\theta,\ 0)$.
- $\hat{\mathbf{z}}_T$: $\hat{\mathbf{x}}_T \times \hat{\mathbf{y}}_T$ (completing the right-handed system).

The axes in world-frame coordinates are:

$$\hat{\mathbf{x}}_T = \mathbf{R}_{\mathbf{u}}(\phi)\begin{pmatrix}0\\0\\1\end{pmatrix} = \begin{pmatrix}\sin\theta\,\sin\phi\\ -\cos\theta\,\sin\phi\\ \cos\phi\end{pmatrix} \tag{7}$$

$$\hat{\mathbf{y}}_T = \mathbf{u} = \begin{pmatrix}\cos\theta\\ \sin\theta\\ 0\end{pmatrix} \tag{8}$$

$$\hat{\mathbf{z}}_T = \hat{\mathbf{x}}_T \times \hat{\mathbf{y}}_T = \begin{pmatrix}-\cos\phi\,\sin\theta\\ \cos\phi\,\cos\theta\\ \sin\phi\end{pmatrix} \tag{9}$$

10.2 Rotation matrix

Assemble the rotation matrix $\mathbf{R}_T$ with these axes as columns (the matrix that maps target-frame coordinates to world-frame coordinates):

$$\mathbf{R}_T = \begin{bmatrix} \hat{\mathbf{x}}_T & \hat{\mathbf{y}}_T & \hat{\mathbf{z}}_T \end{bmatrix} = \begin{bmatrix} \sin\theta\sin\phi & \cos\theta & -\cos\phi\sin\theta \\ -\cos\theta\sin\phi & \sin\theta & \cos\phi\cos\theta \\ \cos\phi & 0 & \sin\phi \end{bmatrix}. \tag{10}$$

For a world-frame vector $\mathbf{w}$, its target-frame representation is $\mathbf{w}_T = \mathbf{R}_T^\top \mathbf{w}$, since $\mathbf{R}_T$ is orthogonal.

The translation vector $-\mathbf{R}_T^\top C$ describes the world origin as seen from the target frame. Its y-component is constant $-r$; its x and z components trace a circle of radius $h+L$ centered at $(L,0)$ in the target xz-plane as ϕ varies (Figure 3):

$$-\mathbf{R}_T^\top C = \begin{pmatrix} L-(h+L)\cos\phi \\ -r \\ -(h+L)\sin\phi \end{pmatrix}.$$

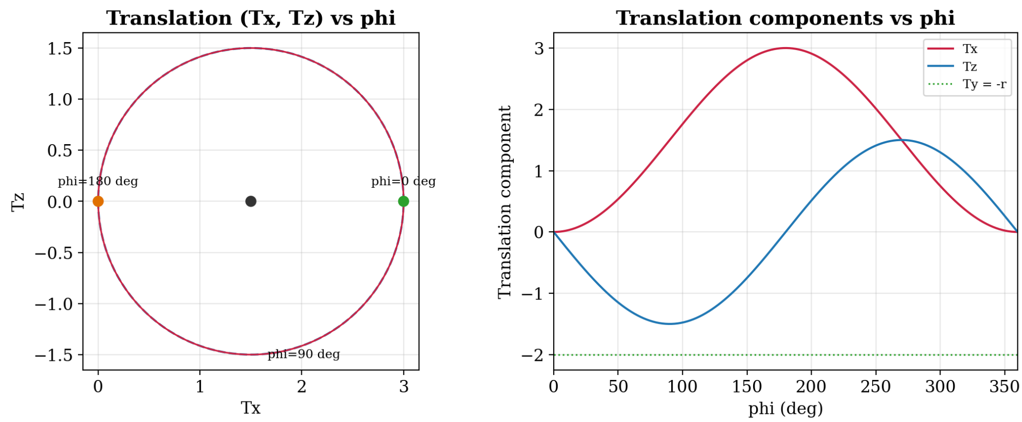


Figure 3: Translation vector $-\mathbf{R}_T^\top C$ as a function of ϕ ($r=2$, $L=1.5$, $h=0$). Left: trajectory in the target xz-plane (a circle of radius $h+L$ centered at $(L,0)$). Right: components vs. ϕ.

10.3 Translation component

The target origin in world coordinates is $C(\theta,\phi)$. A point $\mathbf{p}$ in world coordinates maps to target coordinates as:

$$\mathbf{p}_T = \mathbf{R}_T^\top\big(\mathbf{p} - C(\theta,\phi)\big).$$

Computing $\mathbf{R}_T^\top C(\theta,\phi)$:

$$
\begin{aligned}
\mathbf{R}_T^\top C &= \begin{bmatrix} \sin\theta\sin\phi & -\cos\theta\sin\phi & \cos\phi \\ \cos\theta & \sin\theta & 0 \\ -\cos\phi\sin\theta & \cos\phi\cos\theta & \sin\phi \end{bmatrix} \begin{pmatrix} r\cos\theta - L\sin\theta\sin\phi \\ r\sin\theta + L\cos\theta\sin\phi \\ h + L(1-\cos\phi) \end{pmatrix} \\
&= \begin{pmatrix} h\cos\phi + L(\cos\phi - 1) \\ r \\ (h+L)\sin\phi \end{pmatrix}.
\end{aligned}
\tag{11}
$$

10.4 Homogeneous transformation matrix

The 4×4 homogeneous transformation matrix ${}^{\text{world}}\mathbf{T}_{\text{target}}$ maps world-frame coordinates to target-frame coordinates:

$$
\boxed{{}^{\text{world}}\mathbf{T}_{\text{target}} = \begin{bmatrix} \mathbf{R}_T^\top & -\mathbf{R}_T^\top C \\ \mathbf{0}_{1\times3} & 1 \end{bmatrix} = \left[\begin{array}{ccc|c} \sin\theta\sin\phi & -\cos\theta\sin\phi & \cos\phi & L-(h+L)\cos\phi \\ \cos\theta & \sin\theta & 0 & -r \\ -\cos\phi\sin\theta & \cos\phi\cos\theta & \sin\phi & -(h+L)\sin\phi \\ \hline 0 & 0 & 0 & 1 \end{array}\right].}
\tag{12}
$$

10.5 Verification

- The origin C maps to $(0,0,0)$: $\mathbf{R}_T^\top(C - C) = \mathbf{0}$.
- The bottom endpoint $B = C - L\,\hat{\mathbf{x}}_T$ maps to $(-L, 0, 0)$.
- The top endpoint $T = C + L\,\hat{\mathbf{x}}_T$ maps to $(L, 0, 0)$.
- A point $C + \hat{\mathbf{y}}_T$ (one unit along the rotation axis) maps to $(0, 1, 0)$.
- At $\phi = 0$: $\mathbf{R}_T = \begin{bmatrix} \mathbf{e}_z & \mathbf{u} & \mathbf{e}_z \times \mathbf{u} \end{bmatrix}$ with origin at $P(\theta)$; the z-axis is aligned with the segment.